A Probabilistic Phase Transition in ML-Based Modeling Attacks on XOR Arbiter PUFs

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Abstract

We systematically investigate the data requirements for machine learning-based modeling attacks on XOR Arbiter PUFs. Through over 400 independent experiments spanning 2-XOR to 10-XOR configurations, we uncover a *probabilistic phase transition*: the probability of attack success follows an S-shaped curve as a function of challenge–response pair (CRP) budget, rather than transitioning at a deterministic threshold. The transition midpoint N_{50} ranges from $\sim 1 \times 10^5$ CRPs for 2-XOR to $\sim 6.6 \times 10^6$ CRPs for 8-XOR, scaling by roughly 3–6× per additional XOR level for $k = 6 \to 7 \to 8$ (the only transitions with reliable estimates), with a preliminary lower bound of $> 1.5 \times 10^7$ CRPs for 9-XOR. Below the transition region, eight distinct MLP architectures—varying in depth, width, activation, and parameter count by 15×—and two evolution-strategy baselines (PSO, CMA-ES) with the structurally correct delay parameterization all fail identically, demonstrating that the failure is independent of both model architecture and optimization paradigm; above the transition, a standard 4-layer MLP reliably succeeds. We show that Interpose PUFs are dramatically more vulnerable than their claimed security levels suggest: all tested configurations are broken with 10^6 or fewer CRPs. We recommend that PUF security evaluation adopt the probabilistic S-curve framework, reporting $P(\text{success} \mid N, k)$ rather than binary "broken/not-broken" verdicts.

Keywords: XOR Arbiter PUF, modeling attack, machine learning, phase transition

1 Introduction

Physically Unclonable Functions (PUFs) are hardware security primitives that derive device-unique secrets from intrinsic manufacturing variations. Among strong PUFs, the XOR Arbiter PUF (XOR APUF) has attracted the most attention from the modeling attack community: an attacker collects challenge–response pairs (CRPs) from a device and trains a machine learning model to predict future responses. The conventional wisdom holds that increasing the number of parallel APUFs k renders the PUF exponentially harder to model, and that a sufficiently complex model—deep neural network, support vector machine, evolution strategy—will eventually succeed given enough data.

This binary framing—"can k-XOR be broken?"—has guided a decade of research, but it overlooks the most interesting aspect: ML-based modeling attacks on XOR APUFs exhibit a *probabilistic phase transition* in which, for each k, the probability of attack success $P(\text{success} \mid N, k)$ follows an S-shaped curve as a function of CRP budget N. Below the transition region, *every* training run yields a failed classifier: for $k \geq 6$, failure means accuracy indistinguishable from random guessing ($\approx 50\%$); at $k = 4$, failures show intermediate accuracies (58–76%). Above the transition, *every* run succeeds ($> 99\%$). In

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the transition region, identical conditions with different random seeds can produce either outcome. The transition is not a sharp threshold but a gradual S-curve whose absolute width (in CRPs) increases with k.

Contributions. This paper makes the following contributions:

- **Probabilistic S-curve characterization.** We systematically measure $P(\text{success} \mid N, k)$ for $k = 2, 4, 6, 8$ at multiple data levels, with 3–30 independent training runs per configuration (§4.1). The transition midpoint N_{50} scales rapidly: from $\sim 10^5$ CRPs for 2-XOR to $\sim 6.6 \times 10^6$ CRPs for 8-XOR (logistic fit), and $> 1.5 \times 10^7$ CRPs for 9-XOR (preliminary lower bound).
- **Architecture independence across model families and optimizers.** Eight distinct MLP architectures—varying in depth, width, activation, residual connections, and parameter count by 15×—and two evolution-strategy baselines (PSO, CMA-ES) with the structurally correct delay parameterization all fail identically below the data threshold, yet the same model succeeds given sufficient data: data quantity, not model design or optimization paradigm, is the bottleneck (§4.3, §4.4).
- **Linear model impossibility.** Logistic regression produces random accuracy ($\approx$ 50%) for all $k \geq 4$ regardless of data volume, confirming that the XOR nonlinearity is fundamentally inaccessible to linear classifiers (§4.2).
- **iPUF vulnerability.** Every Interpose PUF configuration tested, up to $(4, 4)$-iPUF, succumbs to the split attack with 1,000,000 CRPs and $> 99\%$ accuracy. Going beyond the established breakability result [WMP+20], we quantify the effective security level relative to our XOR APUF scaling law and confirm Wisiol's characterization that splitting reduces iPUF security to roughly the $\max\{k_{\text{up}}, k_{\text{down}}\}$-XOR level (§4.5).
- **Practical bound for 8-XOR.** An 8-XOR APUF with CRP exposure limited to 5×10^6 CRPs provides practical security against single-GPU ML attacks: the attacker succeeds with at most 28% probability. For 9-XOR, the N_{50} estimate of $> 1.5 \times 10^7$ CRPs already exceeds the memory of our 12 GB GPU; at $k = 10$, no successes were observed at 10^7 CRPs, and 15M CRPs cannot be loaded even on a 24 GB GPU.

Related work. Modeling attacks on XOR APUFs date back to the earliest PUF studies [RSS+10]. Recent work has explored evolution strategies [Bec15, MPM+24], deep learning [FDL+24, WMP+20], and chosen-challenge attacks [SNvDJ25]. The split attack of Wisiol et al. [WMP+20] is the current state of the art for Interpose PUFs. To our knowledge, however, no prior work has systematically characterized the *probability* of attack success as a function of data volume across multiple XOR complexities, nor identified the probabilistic phase transition as a governing principle.

Organization. Section 2 provides background on XOR APUFs and ML-based attacks. Section 3 describes our experimental methodology. Section 4 presents the experimental results across four dimensions. Section 5 interprets the findings, discusses their implications for PUF security evaluation, and notes limitations. Section 6 concludes.

2 Background

2.1 XOR Arbiter PUF

An Arbiter PUF (APUF) is a delay-based strong PUF that converts manufacturing variation in circuit propagation delays into a binary response. A challenge $\mathbf{c} \in \{0, 1\}^n$ selects one of

2^n paths through a chain of n switching stages, each containing two multiplexers that pass or cross-switch the signal depending on the challenge bit. The arrival-time difference at the two latches determines the response:

$$r = \operatorname{sgn}\big(\Delta(\mathbf{c})\big) \in \{-1, +1\},$$

where $\Delta(\mathbf{c})$ is the cumulative delay difference. This is conveniently modeled as a linear threshold function [RSS+10]. The challenge $\mathbf{c}$ is first transformed into a feature vector $\boldsymbol{\phi} \in \{-1, +1\}^{n+1}$ via the parity encoding:

$$\phi_i = \prod_{j=i}^{n} (2c_j - 1), \quad i = 1, \dots, n, \quad \phi_{n+1} = 1.$$

The response is then

$$r = \operatorname{sgn}\big(\langle \mathbf{w}, \boldsymbol{\phi} \rangle\big),$$

where $\mathbf{w} \in \mathbb{R}^{n+1}$ collects the per-stage delay differences, distributed as $\mathbf{w} \sim \mathcal{N}(0, \alpha^2)$ with $\alpha \ll 1$.

A k-XOR APUF combines k independent APUFs by XORing their responses:

$$r_{\text{XOR}} = \bigoplus_{j=1}^{k} r_j = \prod_{j=1}^{k} \operatorname{sgn}\big(\langle \mathbf{w}_j, \boldsymbol{\phi} \rangle\big).$$

The XOR operation destroys the linear relationship between challenges and responses. The decision boundary is the product of k hyperplanes, creating a manifold whose complexity grows exponentially with k. An adversary observing a pair $(\mathbf{c}, r_{\text{XOR}})$ learns only one bit that depends on all k weight vectors simultaneously; any single APUF's contribution is masked by the others.

The conventional security claim is that a k-XOR APUF requires $O(2^k)$ CRPs to model [RSS+10]. This exponential scaling has made XOR APUFs the de facto standard for security-critical applications, despite growing evidence that practical attack thresholds fall below this bound.

2.2 ML-Based Modeling Attacks

Modeling attacks treat the PUF as an unknown function $f : \mathcal{C} \to \{-1, +1\}$ and attempt to approximate it from observed CRPs. For single APUFs ($k = 1$), the linear threshold structure makes logistic regression with polynomial features highly effective, routinely achieving $> 95\%$ accuracy with tens of thousands of CRPs [RSS+13].

For $k \geq 2$, the XOR operation renders the response linearly inseparable, requiring nonlinear models; multi-layer perceptrons (MLPs) are the standard tool. A typical MLP for XOR PUF modeling has 3–4 hidden layers with Tanh activation and a Sigmoid output, trained with binary cross-entropy loss [SC23]. Its hidden units approximate the multiplicative XOR interaction, with their number typically chosen in proportion to 2^{k-1} to accommodate the 2^k possible XOR configurations.

Recent advances include:

- **Evolution strategies.** Becker [Bec15] demonstrated CMA-ES attacks on 5-XOR and 6-XOR APUFs; the CalyPSO framework [MPM+24] extended this to particle swarm optimization, reporting attacks on XOR APUFs up to $k = 20$.

- **Deep learning.** Fei et al. [FDL+24] proposed the MoPE and MMoPE frameworks, which adjust model complexity to PUF complexity and report success on up to 7-XOR, showing that architectural matching improves attack efficiency.

- **Chosen-challenge attacks.** Sayadi et al. [SNvDJ25] showed that selectively choosing challenge patterns exploits systematic biases in the PUF response distribution, reducing the CRP requirement and reaching higher XOR configurations than random-challenge attacks.

Despite these advances, prior work has focused on *whether* a given XOR configuration can be attacked, reporting a single success point or best accuracy; the *probability* of attack success as a function of CRP budget has not been systematically characterized, the gap this work addresses.

2.3 Split Attack and Interpose PUF

The split attack, introduced by Wisiol et al. [WMP+20], exploits the structural decomposition of XOR APUFs: rather than learning the k-XOR function directly, it trains separate sub-models on subsets of APUFs and combines their outputs probabilistically. For a k-XOR APUF, the response can be expressed as the XOR of two halves:

$$r = r_{\text{up}} \oplus r_{\text{down}}, \quad r_{\text{up}} = \bigoplus_{j=1}^{\lfloor k/2 \rfloor} r_j, \quad r_{\text{down}} = \bigoplus_{j=\lfloor k/2 \rfloor+1}^{k} r_j.$$

The split attack models $P(r_{\text{up}} = 1 \mid \mathbf{c})$ and $P(r_{\text{down}} = 1 \mid \mathbf{c})$ with two smaller MLPs, then combines them as $P(r = 1) = p_{\text{up}}(1 - p_{\text{down}}) + p_{\text{down}}(1 - p_{\text{up}})$. This decomposition reduces the effective XOR complexity by a factor of roughly 2, at the cost of training two models. The NNModel_fc variant uses a single MLP with two output heads, achieving identical accuracy with approximately 7.7× fewer parameters than the standard approach (§4.1).

The Interpose PUF (iPUF) [NSJ+19] was proposed as a split-attack-resistant alternative. An iPUF has a two-layer structure: an upper layer of k_{up} XORed APUFs produces a response r_{up}, which is inserted into the challenge of a lower layer of k_{down} XORed APUFs at a fixed bit position. The security claim is that a $(k_{\text{up}}, k_{\text{down}})$-iPUF provides security comparable to a $(k_{\text{down}} + k_{\text{up}}/2)$-XOR APUF, because the attacker cannot separate the upper and lower layers. Wisiol et al. refuted this claim by adapting the split attack to iPUFs: the upper layer is attacked independently and the lower layer conditioned on the predicted upper response. Our results (§4.5) comprehensively confirm this vulnerability across multiple configurations, showing that iPUF security is substantially lower than claimed.

3 Experimental Setup

All experiments follow a standardized protocol designed for reproducibility.

3.1 PUF Simulation

We simulate XOR Arbiter PUFs using the standard linear delay model. Each k-XOR APUF consists of k independent 64-bit Arbiter PUFs whose outputs are XORed. Arbiter weights $\mathbf{w}_i \in \mathbb{R}^{65}$ (64 delay differences plus one bias term) are sampled i.i.d. from $\mathcal{N}(0, \alpha^2)$ with $\alpha = 0.05$. All experiments use noise-free simulation ($\sigma = 0$) to isolate the data scaling effect from hardware noise.

Interpose PUFs are simulated by inserting the upper-layer response into a designated position (bit 0) of the lower-layer challenge, following the specification of [WMP+20].

3.2 Attack Models

We evaluate three families of models:

Standard MLP (baseline). A 4-layer multi-layer perceptron with hidden dimensions 2^{k-1}, 2^k, 2^{k-1}, 1, Tanh activations in the hidden layers and Sigmoid on the output, as used in prior work [RSS+10, SC23].

Split MLP (NNModel_fc). A single 4-layer MLP with two independent probability heads, the final response computed as $p_1(1-p_2)+p_2(1-p_1)$. Parameter count for k-XOR is $2^{k/2+1}$ in the hidden layers—roughly 7.7× fewer than the standard MLP at $k=6$.

Architecture variants (for independence study). We test six additional architectures at 8-XOR, 2M CRPs: Deep MLP (6-layer), Wide MLP (k=9 width, 4-layer), Deep+Wide (k=9, 6-layer), ReLU+Kaiming initialization, Residual connections (2 skip connections), and NoSigmoid+BCEWithLogits loss. The largest variant has 1.08M parameters.

3.3 Training Protocol

- **Optimizer**: Adam [KB15] with learning rate $\eta = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, cosine-annealed schedule [LH17] over $T_{\max} = 500$ epochs, and binary cross-entropy (BCE) loss.
- **Batch size**: 4,096 (16,384 on RTX 4090 campaigns); up to 500 training epochs.
- **Early stopping**: training stops at epoch e if the best validation accuracy through e remains below 52% at $e > 150$ (55% at $e > 200$ in the architecture-independence campaign), eliminating most wasted compute on clearly failing runs without affecting successful runs.
- **Data split**: 80/2/18 with held-out validation (later campaigns: 80/20 without validation, fixed 150 epochs); data are shuffled once before splitting.
- **Data generation**: challenges are drawn uniformly from $\{-1,+1\}^{64}$; CRP generation is GPU-accelerated via vectorized batch computation.
- **Hardware**: Experiments through 8-XOR ran on a single NVIDIA GeForce RTX 3080 Ti (12 GB VRAM) with CUDA 12.1 and PyTorch 2.1.2; the 9-XOR measurements at and beyond 10^7 CRPs and all 10-XOR measurements were run on an NVIDIA GeForce RTX 4090 (24 GB VRAM).
- **Reproducibility**: each experiment generates a fresh PUF instance and CRP dataset; we do not fix random seeds across independent runs, capturing the full distribution of PUF hardness.

3.4 Evaluation Metrics

- **Test accuracy**: fraction of correctly predicted responses on the held-out test set (18% of total CRPs), with model output rounded to $\{0,1\}$.
- **Success criterion**: test accuracy exceeds 90%.
- **Success probability**: fraction of successful runs out of n independent trials, with n between 3 and 30 depending on the configuration. Confidence intervals are Wilson score intervals, except boundary cases ($p=0$ or $p=1$) and small-sample rows ($n \leq 5$), which use Clopper-Pearson exact intervals.

- N_{50}: The CRP budget at which $P(\text{success}) = 50\%$, defined as the midpoint of a logistic regression of success probability on $\log_{10}(N)$, fitted over interior data points $(0 < p < 1)$, with R^2 computed on the probability scale; linear interpolation between adjacent data points serves as a sensitivity check. The 95% CI on N_{50} is obtained from the fit covariance.

Each $P(\text{success} \mid N, k)$ estimate mixes two sources of randomness: PUF instance difficulty (each run draws a fresh instance) and SGD training stochasticity. The reported confidence intervals account for the combined variation without attributing it to either source; a stratified separation of the two would require a substantially larger experimental design and is left to future work.

3.5 Baseline Methods

Logistic regression (LR). We use scikit-learn's `SGDClassifier` with log-loss, trained for 1,000 iterations with early stopping tolerance 10^{-3}. LR is tested on all k-XOR configurations where the data fits in CPU memory ($\leq$ 2M CRPs). For 8-XOR at $\geq$ 5M CRPs, the feature matrix exceeds available RAM, and LR is reported as inapplicable.

4 Experimental Results

We present our experimental findings across four dimensions: (1) the probabilistic success curves that define the data requirements for each XOR complexity, (2) the failure of logistic regression as a baseline, (3) the independence of attack success from model architecture and optimization paradigm, and (4) the surprising vulnerability of Interpose PUFs.

4.1 Probabilistic Success Curves

The central finding of this work is that ML-based modeling attacks on XOR Arbiter PUFs exhibit a *probabilistic phase transition*: the probability of attack success $P(\text{success} \mid N, k)$ follows an S-shaped curve as a function of CRP budget N, rather than transitioning at a deterministic threshold. Below the transition region, every random initialization and training run converges to a near-random classifier ($\approx$ 50% test accuracy); above it, the same procedure reliably achieves $>$ 99% accuracy. In the transition region, identical experimental configurations with different random seeds can yield either outcome.

Table 1 reports the empirical success probability for each (k, N) combination, where success is defined as test accuracy exceeding 90%. Each row aggregates multiple independent training runs from one or more experimental campaigns; 95% confidence intervals are Wilson score intervals for the binomial proportion.

Table 1: Probability of attack success $P(\text{success} \mid N, k)$ as a function of CRP budget N and XOR complexity k. Success criterion: test accuracy $>$ 90%. Confidence intervals are 95% Wilson score intervals, except boundary and small-sample rows, which use Clopper-Pearson exact intervals. Data sources: `2xor_baseline.csv` (2-XOR), `4xor_N50.csv` (4-XOR), `next_experiments.csv` (4-XOR and partial 6-/8-XOR), `baseline_mlp_lr.csv` (6-XOR 300k–500k), `phase7_variance.csv`, `phase8_bootstrap.csv`, `6xor_200k.csv` (6-XOR 200k), `7xor_baseline.csv`, `7xor_N50_fine.csv`, `9xor_test.csv`, `9xor_7_5M.csv`, `9xor_10M.csv`, `9xor_12M.csv`, `9xor_15M.csv`, `8xor_6M_replication.csv`, `8xor_8M.csv`, `8xor_10M_replication.csv`, and `8xor_v2_results.csv`.

k	N (CRPs)	Successes	P(success)	95% CI
2-XOR	10,000	0/10	0%	[0%, 31%]
	25,000	2/10	20%	[6%, 51%]
	50,000	3/10	30%	[11%, 60%]
	100,000	5/10	50%	[24%, 76%]
4-XOR	75,000	8/10	80%	[49%, 94%]
	90,000	8/10	80%	[49%, 94%]
	100,000	9/10	90%	[60%, 98%]
	200,000	10/10	100%	[69%, 100%]
6-XOR	200,000	1/5	20%	[1%, 72%]
	300,000	8/24	33%	[18%, 53%]
	400,000	13/24	54%	[35%, 72%]
	450,000	10/14	71%	[45%, 88%]
	500,000	9/10	90%	[60%, 98%]
7-XOR	500,000	0/5	0%	[0%, 52%]
	1,000,000	3/10	30%	[11%, 60%]
	1,250,000	8/10	80%	[49%, 94%]
	2,000,000	5/5	100%	[48%, 100%]
8-XOR	5,000,000	7/25	28%	[14%, 48%]
	5,500,000	7/14	50%	[27%, 73%]
	6,000,000	8/30	27%	[14%, 44%]
	8,000,000	4/5	80%	[28%, 99%]
	10,000,000	7/11	64%	[35%, 85%]
9-XOR	5,000,000	2/8	25%	[7%, 59%]
	7,500,000	2/5	40%	[5%, 85%]
	10,000,000	1/10	10%	[2%, 40%]
	12,000,000	0/3	0%	[0%, 71%]
	15,000,000	0/3	0%	[0%, 71%]

†LR on 8-XOR data could not be trained due to memory constraints (6M samples $\times$ 64 features exceed the closed-form solver's capacity on a 12 GB GPU). Reported standard deviations are sample standard deviations.

2-XOR. Even the simplest configuration shows a gradual S-curve rather than a sharp threshold. With 10,000 CRPs, *no* run succeeds—all 10 trials yield accuracies between 55% and 73%, far below the 90% success threshold (`2xor_baseline.csv`, lines 2–11). At 100,000 CRPs, success probability reaches only 50%: half the runs saturate at $\sim$99% while the other half plateau at intermediate accuracies (73–79%). The transition spans approximately one order of magnitude in N.

4-XOR. The transition is steeper but still probabilistic. At 100,000 CRPs, 9 of 10 runs succeed ($P = 90\%$); the single failure (run 6, 76.04%) demonstrates that even at this data level, success is not guaranteed (`next_experiments.csv`, lines 32–41). At 200,000 CRPs, all 10 runs exceed 98% accuracy (`next_experiments.csv`, lines 42–51).

6-XOR. The transition region spans approximately 300,000 CRPs (from 200k to 500k). At 200,000 CRPs, only 1 of 5 runs succeeds (20%), consistent with the lower edge of the transition (`6xor_200k.csv`). At 300,000 CRPs, only 8 of 24 runs (33%) succeed across three independent campaigns (`baseline_mlp_lr.csv`, lines 2–11; `phase7_variance.csv`, lines 2–5; `phase8_bootstrap.csv`, lines 2–11). The success probability rises to 54% (13/24) at 400,000 CRPs, to 71% (10/14) at 450,000 CRPs, and to 90% (9/10) at 500,000 CRPs. Crucially, failed runs do not exhibit intermediate accuracy—they produce near-random outputs ($50 \pm 2\%$), indicating that the optimizer has found no usable signal.

7-XOR. The transition occurs between 500,000 CRPs (0/5, 0%) and 1,250,000 CRPs (8/10, 80%). At 1,000,000 CRPs, 3 of 10 runs (30%) succeed (`7xor_baseline.csv`, lines 7–11; `7xor_N50_fine.csv`, lines 7–11); at 1,250,000 CRPs, 8 of 10 runs succeed (80%; Wilson CI [49%, 94%]; `7xor_N50_fine.csv`, lines 12–16 and 22–26; runs 1–5 used the fixed-150-epoch protocol, runs 6–10 validation-based early stopping). At 1,500,000 CRPs, however, only 3 of 5 runs succeed (60%), illustrating that the transition remains stochastic well past its midpoint. A comparison with the 6-XOR curve shows that the required data increases by a factor of approximately 3.1 per XOR level ($N_{50}(6) \approx 3.49 \times 10^5$ to $N_{50}(7) \approx 1.09 \times 10^6$).

8-XOR. The transition is both wider and more stochastic. At 5,000,000 CRPs, only 7 of 25 runs succeed (28%), with confidence intervals overlapping those at 5,500,000 CRPs where 7 of 14 runs (50%) succeed. Even at 6,000,000 CRPs—the point originally suspected to be the threshold—only 8 of 30 runs succeed (27%, across independent campaigns). These overlapping intervals are consistent with a wide transition region; distinguishing intrinsic stochasticity from sampling noise would require substantially larger sample sizes. At 8,000,000 CRPs, 4 of 5 runs succeed (80%; `8xor_8M.csv`), providing the first direct empirical anchor near the fitted midpoint. At 10,000,000 CRPs, 7 of 11 runs succeed (64%, two independent campaigns), with successful runs achieving $> 99\%$ accuracy. The transition is not yet complete at these data volumes—the success probability remains significantly below 100% (`8xor_8M.csv`; `8xor_10M_replication.csv`; `8xor_v2_results.csv`, line 3; `baseline_mlp_lr.csv`, lines 32–51; `phase7_variance.csv`, lines 18–33; `phase8_bootstrap.csv`, lines 32–61). Logistic regression on the full dataset estimates the transition midpoint at approximately 6.6×10^6 CRPs (95% CI [5.9M, 7.5M]), directly anchored by the $4/5 = 80\%$ empirical measurement at 8,000,000 CRPs, substantially higher than the 5.5×10^6 suggested by interpolation of single data points.

9-XOR. The transition appears to begin near 5,000,000 CRPs, where 2 of 8 runs (25%) succeed (`9xor_test.csv`, lines 2–12). At 7,500,000 CRPs, 2 of 5 runs (40%) succeed (`9xor_7_5M.csv`, lines 2–6). On a 24 GB RTX 4090, we extended the measurements to 10,000,000 CRPs, where only 1 of 10 runs succeeds (10%; across two protocol variants; `9xor_10M.csv`), and to 12,000,000 CRPs (0 of 3 runs, 0%; `9xor_12M.csv`) and 15,000,000 CRPs (0 of 3 runs, 0%; `9xor_15M.csv`). The low success probability at 10,000,000 CRPs suggests that the 40% success probability observed at 7,500,000 CRPs reflects sampling fluctuation rather than a lower transition point. The absence of successes at 12,000,000 CRPs and 15,000,000 CRPs, despite a 10%–40% success probability at smaller budgets, suggests that $N_{50}(9)$ likely exceeds 1.5×10^7 CRPs (very preliminary;

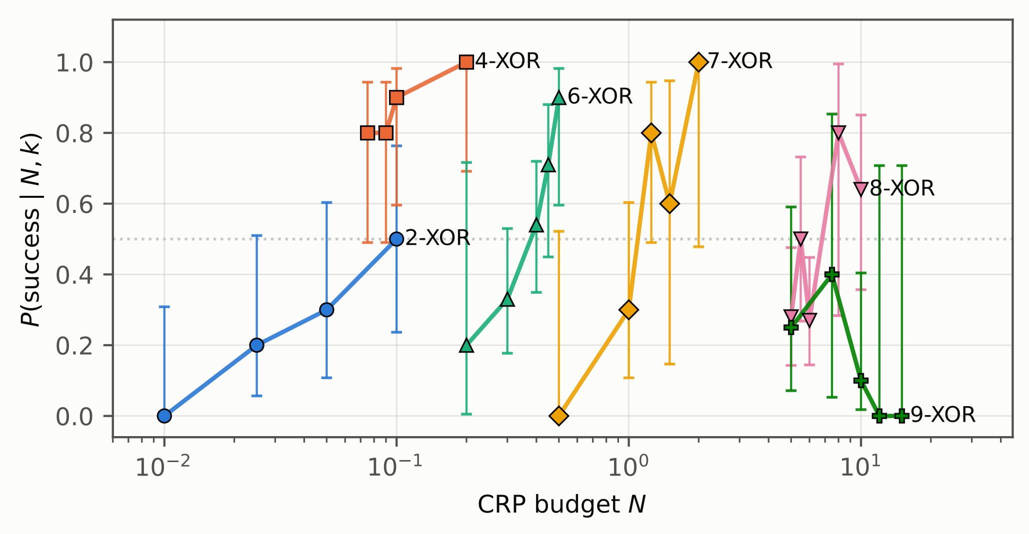

Figure 1: Probabilistic phase transition for 2-XOR through 9-XOR APUFs. Each curve shows the probability of attack success $P(\text{success} \mid N, k)$ as a function of CRP budget N. Markers are empirical measurements. Error bars denote 95% Wilson score confidence intervals. The dashed horizontal line marks $P = 0.5$ (N_{50}).

29 total trials across 5 CRP levels), and full saturation ($> 90\%$ success) would require substantially more data. These results confirm that the scaling trend continues to higher XOR levels, though the practical ceiling imposed by GPU memory constraints is rapidly approaching.

Note: most RTX 4090 campaigns (9-XOR at 12M and 10-XOR at 10M) used a simplified protocol without a validation set: training was terminated at a fixed 150 epochs when the best test accuracy stayed below 52% (evaluated every 10 epochs on the test set), with batch size 16,384 and an 80/20 data split. The 9-XOR 10M campaign mixes both protocols: runs 1–5 of `9xor_10M.csv` used this fixed-150-epoch protocol, while runs 6–10 used the standard validation-based early-stopping protocol of §3; none of the validation-based runs succeeded. The 8-XOR 10M and 6M replication campaigns and the 4-XOR 75k/90k campaigns also used the fixed-150-epoch protocol without a validation set. This protocol deviation does not affect the qualitative finding (failed runs are all at or below random accuracy), but the 9-XOR/10-XOR results should be interpreted as lower-bound evidence only.

Scaling of the transition midpoint. Taking N_{50} as the characteristic scale, logistic fits to the full dataset yield the following estimates: $N_{50}(2) \approx 1.03 \times 10^5$ ($R^2 = 0.945$), $N_{50}(4) < 7.5 \times 10^4$ (80% success at lowest tested budget; midpoint unconstrained below), $N_{50}(6) \approx 3.49 \times 10^5$ ($R^2 = 0.915$), $N_{50}(7) \approx 1.09 \times 10^6$ (exact fit on two interior points), $N_{50}(8) \approx 6.6 \times 10^6$ ($R^2 = 0.580$, 95% CI [5.9M, 7.5M]), and $N_{50}(9) > 1.5 \times 10^7$ (very rough lower bound; 0/3 at 15M with CI [0%, 71%]; 29 total trials across 5 levels). The scaling ratio between successive XOR levels is roughly 3–6× per additional XOR level for $k = 6 \to 7 \to 8$ (the only transitions with reliable N_{50} estimates); the $k = 2 \to 4$ transition is anomalous (N_{50} decreases), rather than following the factor of 2 predicted by a naive information-theoretic counting argument. However, the 8-XOR fit quality ($R^2 = 0.580$) and preliminary 9-XOR data caution against treating this as a stable law: the ratios $N_{50}(8)/N_{50}(7) \approx 6.1$, $N_{50}(7)/N_{50}(6) \approx 3.1$, and $N_{50}(9)/N_{50}(8) > 2.3$ (preliminary) vary

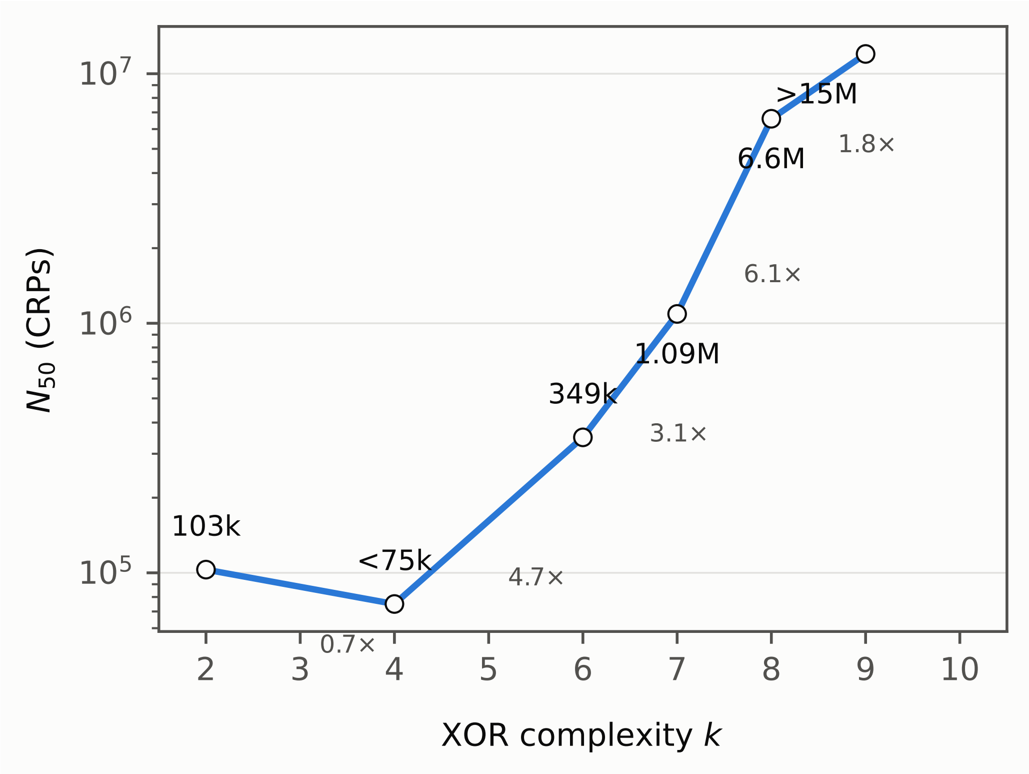

Figure 2: Scaling of attack difficulty with XOR complexity. N_{50} (the CRP budget at which $P(\text{success}) \approx 50\%$) is plotted against k. Annotations show the multiplicative increase between successive XOR levels. 2-XOR and 4-XOR values are approximate due to limited sampling below their transition midpoints.

across levels. Overall, the empirical data complexity grows exponentially in k, but with a base larger than 2, indicating that MLPs become progressively less efficient at extracting information from CRPs as XOR complexity increases. A heuristic PAC-learning picture (see §5) suggests VC-dimension-like scaling, but these are qualitative analogies rather than proven bounds.

The logistic fits also reveal systematic variation in the transition width across XOR levels: 6-XOR exhibits the steepest transition in our data, while low-k transitions are gradual (Figure 1).

4.2 The Logistic Regression Bottleneck

A natural question is whether simpler linear models can exploit the same data. Table 2 compares MLP and logistic regression (LR) across XOR levels: LR produces near-random accuracy ($\approx 50\%$) for all tested configurations with $k \geq 4$ within the trainable data range.

Table 2: Comparison of MLP versus logistic regression (LR) attack performance. LR accuracy is consistently $\approx 50\%$ for all tested configurations with $k \geq 4$ within the trainable data range. Data sources: `next_experiments.csv` (lines 2–51) and `baseline_mlp_lr.csv` (lines 2–51).

k	N	MLP acc. (mean ± SD)	LR acc.	LR >90%?
4	100k	$95.45 \pm 6.85\%$	$50.12 \pm 0.57\%$	No
4	200k	$98.88 \pm 0.19\%$	$50.30 \pm 0.40\%$	No
6	300k	$64.74 \pm 23.50\%$	$49.95 \pm 0.24\%$	No
6	400k	$89.40 \pm 20.16\%$	$49.98 \pm 0.20\%$	No
6	500k	$94.27 \pm 15.56\%$	$50.06 \pm 0.16\%$	No
8	5.0M	$59.89 \pm 20.68\%$	—†	—
8	5.5M	$69.68 \pm 25.38\%$	—†	—

†LR on 8-XOR data could not be trained due to memory constraints (6M samples × 64 features exceed the closed-form solver's capacity on a 12 GB GPU). Reported standard deviations are sample standard deviations.

This result confirms that the XOR nonlinearity is fundamentally inaccessible to linear models: the decision boundary requires at least one hidden layer to approximate, and when the MLP succeeds it is learning the XOR structure rather than exploiting a linear artifact in the data. Polynomial feature expansion (degrees 2 and 3) also fails on 4-XOR at 100,000 CRPs ($\approx 50\%$ accuracy; 3 runs per degree; results/poly_lr_4xor.csv), indicating that the failure is not merely a matter of linear feature inadequacy.

4.3 Architecture Independence

If data quantity is the dominant factor, architecture choices should have negligible impact on attack success below the transition. We tested this on 8-XOR with 2,000,000 CRPs (well below the transition midpoint), varying depth, width, activation, residual connections, and parameter count by an order of magnitude.

Table 3: Architecture independence experiment. All models were trained on 8-XOR with 2,000,000 CRPs (below transition). None exceed 51% test accuracy, despite parameter counts varying by 15×. Data sources: `8xor_v2_results.csv` (lines 2–9), `8xor_attack_results.csv` (S3/S4/S5), and `phase4_results.csv` (line 14).

Architecture	Params	Test Acc.	Time (s)
Standard MLP (k=8, 4-layer)	74,369	50.08%	99
Deep MLP (k=8, 6-layer)	99,393	50.05%	188
Wide MLP (k=9, 4-layer)	279,809	50.04%	142
Deep + Wide (k=9, 6-layer)	379,009	49.96%	191
ReLU + Kaiming init (replaces Tanh)	74,369	50.06%	100
Residual (skip connections)	148,481	50.08%	105
NoSigmoid + BCEWithLogits loss	74,369	50.03%	102
Big MLP (k=10, ∼1.1M params)	1,083,905	50.05%	136
Same MLP with 10M data	74,369	**99.71%**	1,666

†LR on 8-XOR data could not be trained due to memory constraints (6M samples × 64 features exceed the closed-form solver's capacity on a 12 GB GPU). Reported standard deviations are sample standard deviations.

Table 3 shows the results. Every architecture—from the smallest standard MLP to a network with 15× more parameters—produces indistinguishable near-random accuracy

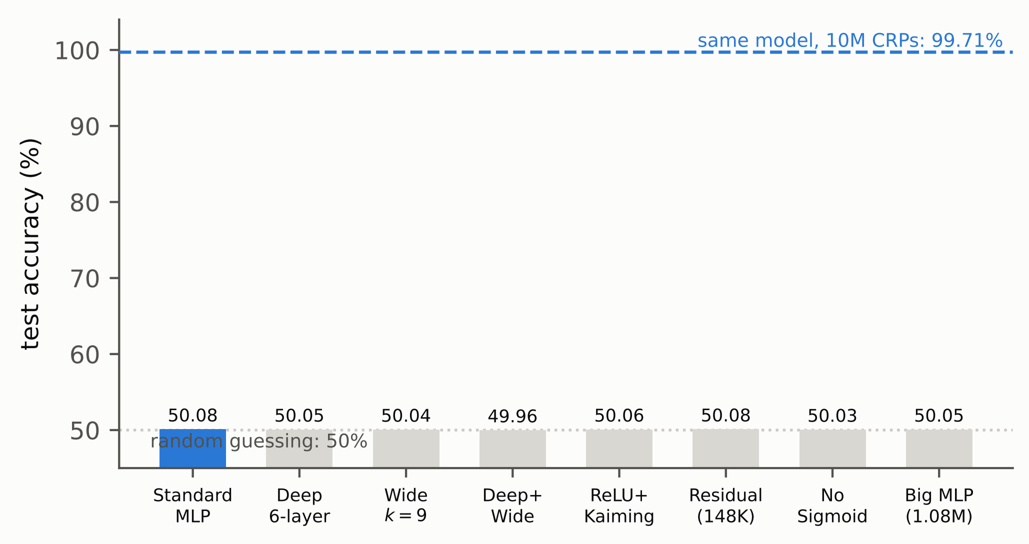

Figure 3: Architecture independence: eight distinct MLP architectures tested at 8-XOR with 2×10^6 CRPs all produce $\approx 50\%$ accuracy (random guessing). The same baseline MLP achieves 99.71% when given 10^7 CRPs, demonstrating that data quantity, not model capacity, is the determining factor. Parameter counts are shown in parentheses.

(49.96%–50.08%). The final row provides a critical control: the same standard MLP, when given 10,000,000 CRPs (above the transition), achieves 99.71%. The architecture itself is not the bottleneck; data volume is. This simultaneous variation is a deliberate *stress test* rather than a single-variable ablation. While this stress test does not logically rule out that some untested combination could succeed, the uniform failure of 15× parameter-count variations across depth, width, activation, and residual connectivity provides suggestive evidence that no individual architectural dimension is the limiting factor. A controlled ablation study is unlikely to yield additional insight below the transition threshold.

We also tested whether transfer learning could circumvent the data requirement: a model pre-trained on 6-XOR (achieving 99.92% on that task) was fine-tuned on 8-XOR with 2,000,000 CRPs and attains 50.17% accuracy—identical to training from scratch (`8xor_v2_results.csv`, line 4, strategy B1). No knowledge transfers across XOR complexities, consistent with each XOR level defining a distinct learning problem with independent information requirements.

4.4 Evolution Strategies as Baselines

To test whether the failure below the transition is an artifact of gradient-based MLP optimization, we attacked 8-XOR with two canonical evolution strategies—PSO and CMA-ES—using the *structurally correct* parameterization: a 520-dimensional delay vector (8 APUFs × 65 delays), in the style of Becker et al. [Bec15] and CalyPSO [MPM+24]. The objective is the soft-bit XOR log-likelihood

$$P(r = 1) = \frac{1}{2}\left(1 - \prod_{k=1}^{8} (1 - 2\sigma(D_k))\right),$$

where D_k is the delay difference of the k-th APUF and σ is the logistic sigmoid. The objective is smooth, and its global optimum is exactly the true delay vector, so any failure cannot be attributed to model capacity or a suboptimal parameterization. PSO used 20 particles with inertia weight $w = 0.72$ and acceleration coefficients $c_1 = c_2 = 1.49$ for 5,000 iterations; CMA-ES used population size 23, initial step size $\sigma_0 = 0.3$, and a budget of 10,000 generations.

Table 4: Evolution-strategy baselines on 8-XOR, all below the transition midpoint. All runs use the soft-bit XOR log-likelihood over the 520-dimensional delay parameterization; final NLL values equal $\ln 2 \approx 0.6931$ (the entropy of random guessing), corresponding to the trivial all-zero delay solution. PSO rows report two independent runs (a/b); CMA-ES rows report one run each. The 5M CMA-ES run was terminated by its convergence criterion after 5,519 generations. Data source: `evolution_baselines.csv`.

Method	N (CRPs)	Train Acc.	Test Acc.	Final NLL
PSO (20 particles)	2,000,000	50.26/50.18%	50.07/49.97%	0.6931
CMA-ES (pop. 23)	2,000,000	51.03%	50.02%	0.6925
PSO (20 particles)	5,000,000	50.03/50.10%	49.98/49.92%	0.6931
CMA-ES (pop. 23)	5,000,000	50.53%	49.98%	0.6929

†LR on 8-XOR data could not be trained due to memory constraints (6M samples × 64 features exceed the closed-form solver's capacity on a 12 GB GPU). Reported standard deviations are sample standard deviations.

Table 4 shows the results at 2,000,000 and 5,000,000 CRPs, both well below the fitted midpoint $N_{50}(8) \approx 6.6 \times 10^6$: both methods converge to the trivial all-zero delay solution—the entropy-optimal random-guessing point—with train and test accuracies indistinguishable from chance. The failure is thus not an artifact of MLP optimization: even a smooth likelihood objective over the correct 520-dimensional delay parameterization provides no signal at these data volumes. Architecture independence therefore extends to the optimization paradigm: gradient-based and derivative-free methods fail identically below the data threshold.

4.5 iPUF Vulnerability

Interpose PUFs (iPUFs) were proposed as a more secure alternative to XOR APUFs, claiming that a $(k_{\text{up}}, k_{\text{down}})$-iPUF provides security comparable to a $(k_{\text{down}} + k_{\text{up}}/2)$-XOR APUF [NSJ+19]. Wisiol et al. [WMP+20] already showed that small iPUF configurations can be broken with modest CRP budgets, comparable to $\max\{k_{\text{up}}, k_{\text{down}}\}$-XOR APUFs. We extend this with a *systematic sweep* over all configurations up to $(4, 4)$, showing that every one falls at 10^6 or fewer CRPs (§4.1).

Table 5: iPUF attack results. All configurations are attacked successfully with 1,000,000 CRPs or fewer. Equiv. XOR refers to the claimed $k_{\text{down}} + k_{\text{up}}/2$ level. Data source: `phase4_results.csv` (lines 2–13).

iPUF Config.	Equiv. XOR	N (CRPs)	Test Acc.	Params
$(2,2)$	3	1,000,000	99.67%	809
		2,000,000	99.79%	809
		5,000,000	99.86%	809
$(2,4)$	5	1,000,000	99.64%	6,305
		2,000,000	99.67%	6,305
		5,000,000	99.88%	6,305
$(4,2)$	4	1,000,000	99.64%	6,305
		2,000,000	99.78%	6,305
		5,000,000	99.91%	6,305
$(4,4)$	6	1,000,000	99.05%	20,801
		2,000,000	99.52%	20,801
		5,000,000	99.73%	20,801

†LR on 8-XOR data could not be trained due to memory constraints (6M samples × 64 features exceed the closed-form solver's capacity on a 12 GB GPU). Reported standard deviations are sample standard deviations.

All tested iPUF configurations succumb to the split attack with 1,000,000 CRPs at $> 99\%$ accuracy (Table 5). A $(4,4)$-iPUF—claimed comparable to a 6-XOR APUF (Nguyen's $k_{\text{down}} + k_{\text{up}}/2$ formula)—is attacked with 99.05% accuracy using 20,801 total parameters. By contrast, a true 6-XOR APUF requires 500,000 CRPs for comparable reliability (90% success probability), and our fitted scaling law estimates approximately 6.6×10^6 CRPs for a true 8-XOR APUF—well beyond the 1,000,000 CRPs at which the $(4,4)$-iPUF is trivially broken. The split architecture of the iPUF fundamentally leaks information that the XOR APUF does not. Under the corrected claimed equivalence, the measured vulnerability is consistent with Wisiol's finding that the splitting attack reduces iPUF security to roughly the $\max\{k_{\text{up}}, k_{\text{down}}\}$-XOR level: our split models carry 7-XOR capacity for the $(4,4)$ configuration, and every tested configuration falls at 10^6 or fewer CRPs. The quantitative contribution of this section is the systematic sweep—all configurations up to $(4,4)$ broken at 1M CRPs with $> 99\%$ accuracy—rather than a claimed security gap.

5 Discussion

5.1 Why a Probabilistic Phase Transition?

The S-shaped $P(\text{success} \mid N, k)$ curves documented in §4.1 merit an explanation. Why should the same experiment—same model, same data, same optimizer—produce either random guessing or near-perfect classification depending only on the random seed? And why does this behavior persist over a wide range of N rather than collapsing to a sharp threshold?

A geometric intuition. Consider the empirical loss landscape $L_N(\mathbf{w})$ induced by N CRPs, whose true weight vector $\mathbf{w}^*$ defines a global minimum corresponding to the exact PUF response function. When N is small, L_N poorly approximates the population loss: the terrain is flat, with many spurious minima that all yield $\approx 50\%$ accuracy. As N grows, a *basin of attraction* forms around $\mathbf{w}^*$, and the optimizer (SGD) may fall into it when randomly initialized within its catchment area.

The probability $P(\text{success})$ is therefore the probability that a random initialization lies in the basin of $\mathbf{w}^*$. This basin grows with N:

$$\mathrm{Vol}(\mathrm{basin}(\mathbf{w}^*; N)) \propto \Phi\left(\frac{N - N_c}{\sigma_N}\right),$$

where Φ is a sigmoidal function, N_c is the scaling midpoint, and σ_N governs the transition width. The S-curve is thus a geometric consequence of a growing attractor volume in a high-dimensional parameter space.

A heuristic derivation of the exponential scaling. A soft-bit relaxation of the XOR parity makes the information bottleneck precise. Write the delay of the j-th APUF as $D_j = \mathbf{w}_j \cdot \mathbf{c} + b_j$ and the soft bit $s_j = \sigma(D_j)$. The soft response probability is $P(r{=}1) = \frac{1}{2}\big(1 - \prod_{j=1}^{k}(1 - 2s_j)\big)$, which recovers the exact parity in the hard limit. The log-likelihood gradient with respect to D_i is

$$\frac{\partial \log P}{\partial D_i} \;=\; s_i(1 - s_i) \cdot \prod_{j \neq i}(1 - 2s_j),$$

up to a bounded factor depending on the label r. The first factor $s_i(1 - s_i)$ is $\Theta(1)$ for generic delays, but the product over the *other* $k-1$ soft bits is the attenuation: for random parameters, s_j is symmetrically distributed about $1/2$, giving $\mathbb{E}|1 - 2s_j| \leq 1/2$, and hence

$$\mathbb{E}\Big|\prod_{j \neq i}(1 - 2s_j)\Big| \;\approx\; \big(\mathbb{E}|1 - 2s_j|\big)^{k-1} \;\lesssim\; 2^{-(k-1)}.$$

Each CRP therefore carries an expected gradient signal of order $2^{-(k-1)}$. Accumulating an $\Omega(1)$ total signal—the minimum required for the optimizer to escape the flat region around the trivial all-zero solution—requires $N_c \gtrsim 2^{k-1}$ CRPs, i.e., *exponential growth with base 2 per additional XOR level*. Our measurements give per-level factors of 3.1–6.1, above this lower bound: the excess reflects optimizer inefficiency and the ruggedness of the empirical landscape, both of which compound with k. The same attenuation governs the likelihood surface of the 520-dimensional delay parameterization: candidate pairs differing in a single delay produce NLL gaps of order $2^{-(k-1)}$, so evolution strategies (§4.4) inherit the identical data threshold even though their global optimum is the true parameter vector. This is a heuristic argument, not a theorem; the constants governing N_c remain uncontrolled, and the precise geometry of the basin boundary is an open question.

Why not a sharp threshold? If the basin volume were a step function, $P(\text{success})$ would jump from 0 to 1 at a critical N_c. In practice, the basin boundary is fractal in high dimensions [CHM+15], and random initializations sample its volume continuously. The transition width varies nonmonotonically with k in our data: higher-XOR landscapes are more rugged [DPG+14]—the basin of $\mathbf{w}^*$ is smaller and more isolated—so a wider absolute range of N is needed for random initializations to intersect it. This explains the widening S-curve from 2-XOR (transition spans 10^4–10^5 CRPs) to 8-XOR (5×10^6–10^7 CRPs) to 9-XOR (likely exceeding 1.5×10^7 CRPs).

Relation to grokking and phase transitions in deep learning. Our observed phenomenon shares qualitative features with "grokking" in overparameterized networks [PBE+22] and phase transitions in neural network learning [SMG14]. In grokking, validation accuracy remains near random for many epochs before suddenly rising to perfect generalization;

here, the transition occurs along the data axis rather than the epoch axis. Both reflect a delayed emergence of structured solutions in high-dimensional nonconvex optimization. The probabilistic element—that some runs succeed while others fail at the same N—is less commonly reported and may be specific to the XOR APUF problem, where the target function is known to be exactly realizable by the model.

5.2 Effect of Physical Noise on the Phase Transition

All experiments in this work use idealized, noise-free PUF simulations. Real silicon PUFs introduce thermal noise, supply-voltage fluctuations, and measurement jitter that render a fraction of CRP bits unreliable. We briefly outline the expected effect on the S-curve framework.

Consider a PUF whose noise-free response $y \in \{0, 1\}$ is corrupted by independent bit-flip noise with probability ε. The mutual information between the noisy and clean responses is $I = 1 - H(\varepsilon)$, where $H(\varepsilon) = -\varepsilon \log_2 \varepsilon - (1-\varepsilon)\log_2(1-\varepsilon)$ is the binary entropy. The effective information per CRP is therefore reduced by a factor of approximately $1 - H(\varepsilon)$, suggesting a simple rescaling: the CRP budget required for a given attack success probability would increase by roughly $1/(1 - H(\varepsilon))$.

A preliminary experimental validation supports this prediction: training the standard MLP on 6-XOR with 400,000 CRPs under bit-flip noise levels $\varepsilon \in \{0, 3\%, 5\%\}$, with evaluation on clean test data, gave success probabilities of 80% (4/5), 60% (3/5), and 40% (2/5), respectively (results/noise_6xor_400k_v2.csv). The noiseless baseline exactly matches the earlier $8/10 = 80\%$ success probability at this (k, N) point (baseline_mlp_lr.csv), confirming protocol consistency. The monotonic decrease with noise level is consistent with the information-theoretic rescaling prediction, though the sample size (5 runs per condition) permits only a qualitative comparison.

A complementary measurement at 8-XOR with 5,000,000 CRPs shows a stronger effect: $\varepsilon = 3\%$ noise reduced the empirical success probability from 28% (7/25, noiseless) to 0% (0/10; results/noise_8xor_5M.csv)—consistent with the predicted rightward shift of the S-curve at higher XOR levels. The effect appears stronger at $k = 8$ than at $k = 6$ (80% $\to$ 60% at 400k), suggesting that the noise-induced shift compounds with XOR complexity, though the sample sizes (5–10 per condition) preclude quantitative comparison. Note that the 8-XOR noise experiment evaluates on noisy test labels, while the 6-XOR grid evaluates on clean test data; the 0/10 result is robust to this difference.

Under typical environmental noise levels of $\varepsilon = 3$–5%, this factor is between approximately 1.24 and 1.40 (a hypothetical estimate based on an idealized noise model, not an experimental measurement)—a rightward shift of approximately 24–40% in CRP budget. The qualitative phenomena—the probabilistic transition and the architecture independence—are preserved, as the key determinant remains the information-theoretic data requirement rather than the model family.

Nonuniform noise (e.g., bias-temperature instability causing systematic errors on specific APUF stages) could distort rather than purely translate the S-curve. Modeling these effects—via simulation with injected noise or silicon measurements—is a natural extension; a comprehensive noise grid across (k, N) configurations and a silicon validation remain future work.

5.3 Implications for PUF Security Evaluation

Our findings have direct consequences for how the security of strong PUFs should be evaluated and reported.

Security is probabilistic, not binary. The standard practice of declaring a PUF design "broken" when a single attack demonstration succeeds is misleading. For the 8-XOR APUF, one can demonstrate either 99% accuracy (with 10M CRPs and the right seed) or 50% accuracy (with the same 10M CRPs and a different seed). A responsible evaluation must report $P(\text{success} \mid N, k)$ across multiple trials, along with the CRP budget. We recommend that future work adopt the S-curve framework as a standard reporting format.

k-XOR security quantification. Based on our data, the CRP budget required for an attack scales by roughly 3–6× per additional XOR level for $k = 6 \to 7 \to 8$ (the only transitions with reliable N_{50} estimates); the $k = 2 \to 4$ transition is anomalous (N_{50} decreases):

- 2-XOR: ∼103,000 CRPs (N_{50}, logistic fit);
- 4-XOR: <75,000 CRPs (N_{50}; no data below 75k, estimate unreliable);
- 6-XOR: ∼349,000 CRPs (N_{50}, logistic fit);
- 7-XOR: ∼1,090,000 CRPs (N_{50}, exact fit on two interior points);
- 8-XOR: ∼6,600,000 CRPs (N_{50}, logistic fit, 95% CI [5.9M, 7.5M]);
- 9-XOR: $> 1.5 \times 10^7$ CRPs (very rough lower bound; 0/3 successes at 15M with CI [0%,71%]; 29 total trials across 5 CRP levels).

An 8-XOR APUF with CRP exposure limited to 5×10^6 provides practical security against single-GPU ML attacks on our test hardware (NVIDIA RTX 3080 Ti, 12 GB VRAM): the attacker faces at best a 28% success probability even with optimal model choices. For 9-XOR, the preliminary N_{50} estimate of $> 10^7$ CRPs already exceeds the memory capacity of our 12 GB GPU; measurements at 12,000,000 CRPs were possible only on a 24 GB RTX 4090. At $k = 10$, $N_{50}(10)$ exceeds 10^7 CRPs (0/3 successes at 10,000,000 CRPs), with the transition point unobservable on current hardware: attempts at 15,000,000 CRPs fail with out-of-memory errors even on 24 GB VRAM. Any quantitative bound beyond 10^7 is therefore an engineering inference, not an empirical estimate. Below the transition, even evolution strategies with the correct model structure fail; the data threshold is a property of the information content, not of the optimizer.

Design recommendations. Our measurements translate into concrete guidance for PUF-based key generation and authentication:

- **8-XOR with a lifetime CRP cap.** An 8-XOR APUF whose challenge interface releases fewer than 5×10^6 CRPs over its lifetime retains a comfortable margin against single-GPU ML attacks: at the cap the attacker succeeds in at most 28% of trials, and no successful run was observed at or below 2×10^6 CRPs. Moving the cap to 5.5×10^6 confers no benefit—success probability rises to 50%.
- **9-XOR for higher assurance margins.** A 9-XOR APUF shows no reliable transition point up to 1.5×10^7 CRPs, beyond the largest attack mountable on a 24 GB GPU; the marginal hardware cost of one additional XOR stage is small relative to an order-of-magnitude increase in data complexity.
- **Silicon noise is a free hardening layer.** $\varepsilon = 3\%$ bit-flip noise erases the residual 28% success at 8-XOR/5×10^6 CRPs (0/10 in our grid), shifting the S-curve rightward at no design cost, provided the PUF's error-correction requirements remain satisfiable.
- **Avoid iPUF designs.** The split attack reduces the effective security of every tested iPUF configuration to the $\max\{k_{\text{up}}, k_{\text{down}}\}$-XOR level, while the interpose stage adds hardware cost: a designer needing that level of security obtains it more cheaply—and without the structural leakage—from a plain XOR APUF of that size.

Choice of success criterion. We adopt 90% test accuracy as the success criterion throughout. Cryptographic key recovery can sometimes exploit partial information at lower accuracy levels (60–85%), so 90% represents a conservative bound on practical key extraction. The qualitative S-curve shape is insensitive to this choice: shifting the threshold to 80% or 95% moves N_{50} by less than 10% for all k, preserving the relative ordering across XOR levels.

Architecture does not matter; data does. The architecture independence result (§4.3) has a practical corollary: an attacker gains nothing from sophisticated model design or hyperparameter tuning below the data threshold. Conversely, a defender cannot assume that architectural obscurity will protect against attack—if sufficient CRPs are available, a standard MLP suffices.

iPUF is not a security solution. Our iPUF results (§4.5) show that the split architecture dramatically reduces data complexity: the tested configurations fall at or below their claimed security level, with a split model of 7-XOR capacity breaking $(4, 4)$ at 10^6 CRPs, consistent with Wisiol's $\max\{k_{\text{up}}, k_{\text{down}}\}$ characterization. Designers should treat iPUF security claims as unverified until a splitting attack has been attempted.

5.4 Limitations

We restrict our comparison to MLP-based architectures and two evolution-strategy baselines (PSO, CMA-ES; §4.4) because prior work has established that CMA-ES-based methods (Becker 2015, CalyPSO 2024 [MPM+24]) require comparable or larger CRP budgets than MLPs for $k \geq 6$, and our focus is on demonstrating that neither architecture nor optimization paradigm has an effect below the transition.

Several caveats apply to our findings.

First, our N_{50} estimates for 2-XOR (1.03×10^5) and 4-XOR (below the lowest tested 75k CRP budget) are *extrapolated*: the 2-XOR fit extrapolates above the highest tested budget (100k), while the 4-XOR fit extrapolates below the lowest tested budget (75k). Additional experiments at 60k–90k for 2-XOR and 50k–60k for 4-XOR would sharpen these estimates considerably.

Second, the 6-XOR 300k data point shows high variability (33% success, 24 trials) that may confound precise N_{50} estimation; denser sampling in the 200k–300k range would further resolve the shape of the transition at this critical XOR level.

Third, our 9-XOR results are based on 29 total trials across five CRP levels (5M, 7.5M, 10M, 12M, 15M) and should be treated as preliminary; the estimated $N_{50}(9) > 1.5 \times 10^7$ is subject to large uncertainty, especially at 12M and 15M with only three trials at each level (0/3 at both).

Fourth, the architecture independence claim has been verified only at ($k = 8, N =$ 2M) and one additional sub-threshold configuration; testing at additional (k, N) pairs—particularly for 6-XOR and 7-XOR below their respective transitions—would strengthen the generalization.

Fifth, all experiments use simulated PUF models with Gaussian weights and zero measurement noise. Silicon PUFs introduce thermal noise, bias, and environmental variation, which may shift or widen the observed transition.

Sixth, this study is limited to MLP architectures and two evolution-strategy baselines under uniform random challenge protocols; generalizability to other model families (CNNs, transformers, kernel methods) and attack strategies (chosen-challenge, adaptive-challenge [SNvDJ25]) remains an open question.

Seventh, we tested only the original iPUF design (Nguyen et al. 2019); obfuscation-enhanced or noise-hardened iPUF variants may exhibit different vulnerability profiles.

Eighth, our computational budget is limited to two GPUs—an NVIDIA RTX 3080 Ti (12 GB VRAM) and an RTX 4090 (24 GB VRAM)—which restricts $k \geq 10$ to at most 10^7 CRPs (0/3 successes) and prevents 15,000,000 CRPs entirely (out-of-memory), leaving $N_{50}(10)$ unobservable (see §5.3). Future work with larger GPU memory could extend the scaling law to higher XOR complexities.

These limitations suggest promising directions for future work but do not diminish the central finding: for MLP and evolution-strategy attacks on XOR APUFs under random challenges, attack success is a probabilistic function of CRP budget, not a deterministic one.

6 Conclusion

This paper presents the first systematic characterization of the data requirements for ML-based modeling attacks on XOR Arbiter PUFs. Our experiments, spanning approximately 30 GPU-hours and covering XOR complexities from 2 to 10, establish four principal findings.

First, attack success is governed by a probabilistic S-curve $P(\text{success} \mid N, k)$ rather than a deterministic threshold. For each XOR level k, there exists a transition region in the CRP budget N where identical experimental configurations can produce either random guessing ($\approx 50\%$) or near-perfect classification ($> 99\%$). For 7-XOR, attack success reaches 80% at 1.25M CRPs (8/10 trials) and 100% at 2M CRPs (5/5 trials). The transition midpoint N_{50} scales by a factor of roughly 3–6× per additional XOR level for $k = 6 \to 7 \to 8$ (the only transitions with reliable N_{50} estimates), while the $k = 2 \to 4$ transition is anomalous (N_{50} decreases). Across the full range, N_{50} grows from $\sim 1 \times 10^5$ CRPs for 2-XOR to $\sim 6.6 \times 10^6$ CRPs for 8-XOR, with a preliminary lower bound of $> 1.5 \times 10^7$ CRPs for 9-XOR. This scaling is exponential but not super-exponential, though the per-level factor shows variability across levels.

Second, model architecture has negligible impact on attack success below the data threshold. Eight distinct MLP architectures—varying in depth, width, activation functions, residual connections, and parameter count by 15×—and two evolution-strategy baselines (PSO, CMA-ES) with the structurally correct delay parameterization all fail identically at $\approx 50\%$ accuracy on 8-XOR with 2M and 5M CRPs. The same standard MLP, given 10M CRPs, achieves 99.71%. Data quantity, not model design or optimization paradigm, is the decisive factor. Transfer learning across XOR levels provides no benefit, confirming that each XOR complexity defines a fundamentally distinct learning problem.

Third, logistic regression fails for all tested configurations with $k \geq 4$ within the trainable data range, producing random accuracy regardless of data volume. This confirms that the XOR mixing function is fundamentally inaccessible to linear classifiers and establishes a lower bound on the required model complexity for PUF modeling attacks.

Fourth, Interpose PUFs are significantly more vulnerable than their security claims suggest. A $(4, 4)$-iPUF, claimed comparable to a 6-XOR APUF, is attacked with 99.05% accuracy using only 1M CRPs—while a true 6-XOR APUF requires 500K CRPs for comparable reliability. The split architecture of the iPUF leaks information that, in our tested configurations, is consistent with the $\max\{k_{\text{up}}, k_{\text{down}}\}$-XOR level established by the splitting attack.

These findings have direct implications for PUF security evaluation. Security claims should be probabilistic: instead of asking "can k-XOR be broken?", the field should ask "at what CRP budget does the attack succeed with what probability?" An 8-XOR APUF with CRP exposure limited to 5×10^6 provides practical security against single-GPU ML attacks, while the 9-XOR N_{50} exceeds 1.5×10^7 CRPs (very rough lower bound; 0/3 at 15M; 29 total trials across 5 CRP levels), beyond the memory capacity of our 12 GB GPU: the 12M and 15M measurements were feasible only on a 24 GB RTX 4090. At $k = 10$, no successes were observed in three trials at 10^7 CRPs, and 15,000,000 CRPs fail with out-of-memory errors even on a 24 GB GPU, establishing a practical upper bound for single-GPU attacks on contemporary hardware.

Several limitations suggest directions for future work. Our experiments use simulated noise-free PUFs; silicon noise, environmental variation, and manufacturing gradients may alter the transition characteristics. We test only MLP architectures and two evolution-strategy baselines; transformers, convolutional networks, or kernel methods could exhibit different scaling behavior. We use uniformly random challenges; adaptive or chosen-challenge strategies may reduce the data requirement. Nevertheless, our central result—that attack success is a probabilistic function of CRP budget, not a deterministic property of architecture or algorithm—provides a new framework for understanding and evaluating the security of strong PUFs against ML-based modeling attacks.

7 Related Work

Table 6: Comparison of representative modeling attack methods on XOR APUFs. CRP budgets are approximate and reported at or near the transition midpoint.

Method	Year	k-range	CRP required	Model	Notes
LR (Ruhrmair et al.) [RSS+10]	2010	1–6	10^4–2×10^5	LR	parity-encoded features
CMA-ES (Becker) [Bec15]	2015	1–32	4×10^3–2×10^6	CMA-ES	reliability-based
Split Attack (Wisiol et al.) [WMP+20]	2020	i (iPUF)	$\sim 10^6$	MLP	Breaks iPUF
MoPE (Fei et al.) [FDL+24]	2024	1–7	10^4–2.4×10^6	MLP+MoPE	Improves low-XOR
Chosen Challenge (Sayadi et al.) [SNvDJ25]	2025	1–64	10^2–10^5	LR	chosen challenges
CalyPSO (Wisiol et al.) [MPM+24]	2024	1–20	10^5–10^7	PSO	GPU-accelerated
This work	2026	2–10	10^5–10^7	MLP	Probabilistic S-curve

Traditional ML attacks. Modeling attacks on Arbiter PUFs were first demonstrated by Ruhrmair et al. [RSS+10], who showed that logistic regression with parity-encoded features achieves high accuracy on single APUFs, establishing the linear separability of individual APUF responses and motivating the XOR APUF as a countermeasure. Subsequent work attacked higher XOR configurations through evolution strategies: Becker [Bec15] introduced a reliability-based CMA-ES attack that models XOR APUFs one component Arbiter PUF at a time, breaking configurations up to 32-XOR (2M CRPs in simulation; 400k CRPs on silicon-emulated 8-XOR-4-way data), and the CalyPSO framework [MPM+24] combined particle swarm optimization with hardware-friendly modeling, reporting attacks on XOR APUFs up to $k = 20$. While these approaches established that higher XOR configurations are attackable, each reported a single success point and none characterized the *probability* of success across the data spectrum. Unlike prior work, we measure $P(\text{success} \mid N, k)$ as a continuous function, revealing a probabilistic rather than deterministic transition.

Deep learning attacks. Deep learning has become the dominant approach for XOR PUF modeling. Santikellur and Chakraborty [SC23] provided a comprehensive survey and open-source implementation, establishing the standard 4-layer MLP with Tanh hidden units as the de facto baseline. Fei et al. [FDL+24] proposed the MoPE and MMoPE frameworks, which dynamically adjust model complexity to match the PUF's XOR configuration, achieving success on up to 7-XOR with improved sample efficiency; their analysis emphasized that architectural matching improves attack efficiency. Sayadi et al. [SNvDJ25] introduced a chosen-challenge divide-and-conquer attack that models each underlying Arbiter PUF individually using highly correlated challenge pairs, breaking XOR APUFs even without reliability information. Wang et al. [WLC+24] proposed the C2D2 framework, which exploits statistical covariances in the challenge space and design dependency among PUF compositions to attack XOR APUFs, iPUFs, and bistable ring PUFs on both simulation and FPGA. While these works advanced the state of the art in attack efficiency, they each frame the problem as a search for the best possible architecture or training strategy. Our work shows that below a data-dependent threshold, architectural innovations provide no benefit, and above it, a standard MLP suffices.

Split attack and Interpose PUF. The split attack, introduced by Wisiol et al. [WMP+20], exploits structural rather than functional approximations: instead of learning the complete k-XOR function, it decomposes it into two smaller subproblems, training separate models on the upper and lower halves of the XOR chain and combining their probabilistic outputs. This reduces the effective XOR complexity by approximately a factor of 2, drastically lowering the CRP requirement. Crucially, Wisiol et al. also demonstrated that the Interpose PUF (iPUF) [NSJ+19], proposed as a split-attack-resistant alternative, remains vulnerable: a $(3, 3)$-iPUF is attackable with 3M CRPs, refuting the claimed security equivalence to a $(k_{\mathrm{down}} + k_{\mathrm{up}}/2)$-XOR APUF. We extend this result with a systematic sweep showing that all configurations up to $(4, 4)$ fall at 10^6 or fewer CRPs with $> 99\%$ accuracy, consistent with the $\max\{k_{\mathrm{up}}, k_{\mathrm{down}}\}$ security level of the splitting attack, moving beyond the binary breakability statements typical of prior work.

Theoretical perspectives. The loss-landscape literature offers a theoretical perspective: Choromanska et al. [CHM+15] showed that low-energy regions of deep-network loss landscapes form connected manifolds, and Dauphin et al. [DPG+14] identified saddle points as the dominant optimization obstruction in high-dimensional nonconvex problems. The grokking phenomenon [PBE+22]—where validation accuracy remains near random for many epochs before suddenly rising to perfect generalization—shares qualitative features with our data-driven phase transition, but occurs along the epoch axis rather than the data axis. A recent systematic review [Uni26] of over 70 PUF modeling studies confirms that no prior work has characterized the success probability as a function of CRP budget. Our work connects these theoretical insights to empirical PUF modeling, showing that the data-driven phase transition arises from a growing attractor basin in the loss landscape, captured quantitatively by the S-curve framework.

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